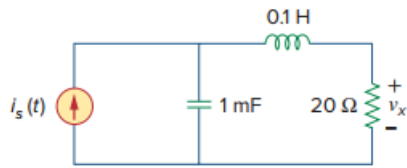


Problem

Determine v_x in the circuit of Fig. 9.57. Let $i_s(t) = 5 \cos(100t + 40^\circ)$ A.

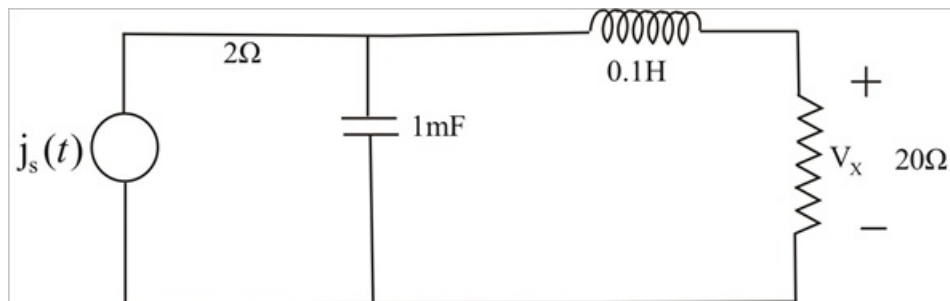
Figure 9.57:



Step-by-step solution

Step 1 of 3

Given $i_s(t) = 5 \cos(100t + 40^\circ)$ Find V_x .



Step 2 of 3

$$\omega = 100$$

$$Z_1 = 20 + j100 \times 0.1$$

$$= 20 + j10$$

$$Z_{eq} = \frac{(20 + j10) \left(\frac{1}{j \times 100 \times 1 \times 10^{-3}} \right)}{20 + j10 + \frac{1}{j100 \times 10^{-3}}}$$

$$= \frac{(20 + j10)(-j10)}{20}$$

$$= \frac{100 - j200}{20} = 5 - j10$$

$$V = I_s Z_{eq}$$

$$= 5 \angle 40^\circ \{11.18 \angle -63.43^\circ\}$$

$$= 55.9 \angle -23.43^\circ$$

Step 3 of 3

$$\begin{aligned}
 V_x &= \frac{20}{20+j \times 100 \times 0.1} V \\
 &= \frac{20}{20+j10} \{55.9 \angle -23.43\} \\
 &= \frac{20(20-j10)}{500} \{55.9 \angle -23.43\} \\
 &= \frac{400-j200}{500} \{55.9 \angle -23.43\} \\
 V_x &= (0.89 \angle -26.56^\circ)(53.9 \angle -23.43) \\
 V_x &= 49.75 \angle -49.99^\circ
 \end{aligned}$$

In time domain,

$$V_x = 49.75 \cos(100t - 49.99^\circ) \text{ .}$$